

# If People Were Data: How to Think About Modern Migration Infrastructure

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Some of you may be aware of this: the title of this talk is referring to a piece by Robert Goodin titled 'If people were money...' from 35 years ago. That paper is about the moral (or ethical, if you prefer, I use both terms interchangeably in this talk) tension between how easy it is for money to cross borders to wealthy countries and how difficult it is, in contrast, for labour migrants.

Nowadays, migration for humans is still difficult and it looks like it has become more difficult for a lot of people, including for a lot of those who would qualify as genuine asylum seekers.

One reason for this, and that is the reason I focus on today, is the rapid development of digital technologies for data capturing and processing, as well as the development of algorithms that allow for assessing large amounts of data in very little time. We are all aware of this and there is plenty of literature about the impact of various dimensions of digital technologies on people's lives as consumers, as citizens, as medical patients etc.

So it is not very surprising that digital technologies to gather and process data are also becoming more relevant when we think about how borders are secured and how migration is managed by receiving countries.

I should note that there is a distinct difference between the implication of Goodin's original title and my remix of it: 'If people were money' is, of course, expected to be completed with something 'they could cross borders without effort or harm'.

It is not quite as straightforward for 'If people were data'. As we will see, for modern border and immigration management, people *are* effectively data in regard to many relevant workflows. And it is not obvious whether this is a good or a bad thing or a bit of both — certainly, with an increase of nationalist tendencies globally and attempts in many wealthy receiving countries to appease right-wing populist movements, the datafication of migrants makes it harder to enter potential receiving countries.

This does not necessarily have to be that way and there are a lot of possible applications of digital technology that could offer real support for migrants. But currently — at least as it looks to me — being data makes it more difficult for many people to migrate from one country to another.

# Digital Data and Migration

Why does the use of digital data make migration more difficult for a lot of people?

A lot of the technology used in border contexts is used for data-driven surveillance. The periphery devices are automated electronic passport gates, autonomous border surveillance towers, drones, balloons, or even satellites. These are, depending on their specific purpose, equipped with biometric image recognition software, infrared vision, radar, CO2 and heartbeat sensors, and whatever other sensor technologies were still science fiction ten years ago.

However, just having these periphery devices collecting impressions of their environment by themselves would not be worth very much. Neither would it help if we needed one actual person for the purpose of monitoring per periphery device, or per a small range of them — that would not be economically attractive and it would also not be possible to harness the full potential of this sort of multi-sensory and in large parts asynchronous surveillance.

So what we need is, of course, systems that store and channel the data from all these different periphery devices, that can label and contextualise them, and that make them searchable and easily accessible and comparable for human operators. And, more recently, they should also include self-learning algorithms that can pre-process the data and make reasonable recommendations to human operators. It is all of these different layers together, networked periphery devices, storage systems, processing units, interfaces between them, and machine learning algorithms connected to these structures that I refer to as *digital migration infrastructure* here.

This is not necessarily a complete definition — perhaps one could argue for a variant that includes the mobile networks used by migrants, for example. Perhaps they are another dimension of digital migration infrastructure but not one I will talk about right now.

My impression is that thinking about digital technology *generally* in the context of migration control and management is a perspective that we encounter more often in the academic literature these days; at the same time, looking at it from an *infrastructure perspective* is still not very common.

There are *some* authors who actually use an infrastructural perspective on migration such as Huub Dijstelbloom ('Borders as Infrastructure: The Technopolitics of Border Control') or Kaelynn Narita ('An infrastructural approach to the *digital* Hostile Environment'). You may be familiar with their work or you will read some of it later and remember this talk.

So perhaps I should briefly highlight that my infrastructure-focused approach today is in contrast almost minimal and very analytical and technical. Perhaps it lacks features that some people would say are the genuinely philosophical ways of

thinking about such technologies. That means, for example, that I do not want to make any strong ontological claims about digital migration infrastructure, or discuss the phenomenology of border surveillance, or whether terms like ‘infrastructural border violence’ (Dijstelbloem 2021) are meaningful (I don’t think so).

We also should be aware that we don’t have to re-invent the wheel when thinking about digital migration infrastructure. A lot of the normative challenges we can find here have already been identified and discussed in domestic digital infrastructure contexts. Often, migrants face similar challenges when engaging with digital migration infrastructure but, of course, the normative relation between them and this infrastructure, and how to evaluate their standing and their options, may — or may not — differ significantly from similar domestic cases, depending on our theoretical starting point (e.g., liberal nationalists will argue that a lot of rights citizens have cannot be held by migrants, while cosmopolitans will disagree).

If we compare digital *migration* infrastructure with *domestic* digital infrastructure, we see that migration infrastructure is (as many other types of digital infrastructure) a site of *dataveillance*, data-based surveillance, as defined by Roger Clarke in 1988 (Clarke 1988): ‘the systematic use of personal data systems in the investigation or monitoring of the actions or communications of one or more persons.’ And for the last ten to fifteen years, there is a healthy academic debate about risks of dataveillance, whether under this label or not, in democratic societies, often in conjunction with the phenomenon of *datafication*, meaning representing people in a specific social role (e.g., as citizens) as an aggregation of data for decision making (see, e.g., Büchi et al. 2022; Van Dijck 2014). Increasingly, this also comes more into focus in philosophical and ethical assessments of migration practices and policies (see, e.g., Meneses Gutierrez 2023).

## Thinking about Infrastructure

If there already is quite a bit of literature on digital technology and migration and most of it does not care about the digital infrastructure aspect at all, why should we think about infrastructure? In order to merit serious consideration, looking at the challenges of technology and migration through an infrastructural lens must have some sort of advantage, otherwise it would just mean going through unnecessary additional motions.

I think the benefit is that not only focusing on single technological components but indeed on a network of different systems, an infrastructure, makes us more aware of the distributed nature of modern border management. It highlights that the central normative challenges are not, for example, that the same drones used for border surveillance can and are also used as military weapons. Instead, it reminds us to also look beyond what we may think of as an immediate object of moral concern to see what context it is embedded in. It may well be that it is this context that should be the reason for our moral concerns, not the periphery device.

## Responsibility Gaps of Various Kinds

One thing that can come to mind, for example, and that has been discussed in philosophy of technology for a while already is responsibility (or accountability). People have been talking about the so-called *responsibility gap* with regard to self-learning machines for more than twenty years now (starting with ‘The Responsibility Gap: Ascribing Responsibility for the Actions of Learning Automata’ by Andreas Matthias (Matthias 2004)). The term *responsibility gap* refers to exactly what the name says, the fact that in some cases of technology use, we can no longer trace some outcomes back to an agent that can be clearly held responsible for it.

What does that mean? In easy cases of tool use, we usually have a fairly clear idea who is responsible for the outcome. If I hit someone on the head with a hammer and that person is severely injured or killed, it is quite clear that I am responsible — not the hammer, not the manufacturer of the hammer. The same goes for more complex machines, for example old-fashioned cars where drivers are in complete control of the vehicle. If I hit someone with my car, I am first and foremost responsible, not the car, not the car manufacturer. There may be circumstances under which I may not be responsible, e.g., if the mechanic who fixed my brakes recently made a mistake and things would have worked out well if only the brakes would have worked properly. In this instance, however, responsibility can be traced to the mechanic who was supposed to fix my brakes.

The original presentation of the responsibility gap was concerned with self-learning algorithms and that outcomes cannot be determined anymore by people setting up or deploying such algorithms — and looking at modern Large Language Models, this is certainly a concern we understand very well, perhaps even better, today. The answer to the question of who is responsible for a certain outcome in these cases remains often unclear, and must, at the end of the day, likely include questions about who (if anyone) counts as an actor in causal chains leading to specific outcomes (see, e.g., Carter 2026).

There has also been more work in recent years to analyse what we are talking about when we talk about responsibility gaps, as we might refer to different types of this phenomenon. One can, for example, distinguish between gaps regarding *culpability*, *moral accountability*, *public accountability*, or *active responsibility* (see Santoni De Sio and Mecacci 2021), and probably there are still more types and categories that become relevant here, depending on practical context and theoretical approach.

This problem is also not unique to cases of self-learning machines, although these particular cases come with their very own particular technical context. We have similar questions with organisations, corporations, and in other contexts where large numbers of people work with each other in a coordinated way. We can get outcomes where none of the people involved are fully responsible (if at all), which has been labelled *the Problem of Many Hands* by Dennis Thompson almost half a

century ago (see Thompson 1980). Philosophers and ethicists have been thinking about group responsibility and group ethics for a while now.

One of the main differences between these two ethical spheres appears to be that in one instance, we are thinking about cases where responsibility gaps can occur in processes that only involve human agents (who can, at least potentially, bear responsibility, partial or full) while in the other instance, we have algorithms as artificial agents that cannot (at least that's the view of most philosophers and ethicists as far as I am aware) be held responsible.

With digital migration infrastructure, as well as with other digital infrastructure that is used to administrate and manage people, we have a complex mixture of both: first we have a lot of people interacting with connected systems and applications from different places, at different times, often knowing only as much as they need to know for their particular tasks but little about the tasks of other people within this infrastructure. Second, we likely have algorithms that, in the widest sense, recommend something — perhaps simply how data should be sorted or filtered, or perhaps less simply, how much of a potential threat a specific person may be.

It seems that data processing infrastructure, particularly when it includes algorithmic intelligence, has a big potential to create responsibility gaps of various kinds — old-fashioned ones and new ones.

Usually, we want to be able to clearly assign responsibility (or accountability or something similar) which means a lot of effort in the literature goes into finding possible ways to bridge or plug the responsibility gap. However, as John Danaher points out, responsibility gaps in automated systems also allow human agents to unload responsibility in the case of tragic choices (see Danaher 2022) — for example, medical triage situations when only limited resources are available. Similar psychological mechanisms may make it easier for border and immigration staff to work along political guidance on migration they may be morally uncertain of, if decisions to reject a potential immigrant are already recommended by an automated system.

By adopting an infrastructural view, we get a better feel for the complexity of the technological aspects of a particular problem or phenomenon we are analysing. We can understand better why existing digital systems for migration control and management are often flawed and prone to make certain kinds of errors or leave room for such errors — because they are complex, made up from various modules and components with different purposes within the system.

Due to this complexity, it may sometimes be challenging to clearly see what relevant normative questions there are and what details are relevant when examining or trying to answer these questions.

One methodological approach that can be helpful here is the input-process-output model.

# The Input-Process-Output Model (IPO)

Thinking about ethical and political evaluation of digital technology from an infrastructure perspective can be a demanding task. The questions we face are complex and often involve fields of expertise that we don't feel sufficiently familiar with (if we are working on our own) or which need to be assigned to the right people (if we are working in a team).

If we think about the normative challenges of digital technology and migration as primarily driven by digital infrastructure developments, thinking of the questions involved through the lens of the Input-Process-Output (or IPO) model can be helpful to get a better idea what sort of agents, data, and technical components are involved and which of these are particularly relevant for our questions.

The IPO model is used in Computer Science for software engineering and, more generally in Systems Science, for process and systems analysis (often also in business contexts). The distinction of three different categories can help us to focus on one particular aspect of the model to gain a better understanding of what happens or needs to be done.

This is relevant, e.g., in software engineering if we know what sort of input a piece of software will be getting and what sort of output it is expected to generate. Knowing these two things allows us to decide what the software will have to do in order to get from the expected input state to the expected output state.

The idea of applying the IPO model to normative questions is not new, of course, and I came across this recently in an article by Sung Jun Han (Han 2025) from Vanderbilt University who applies this to questions of the use of Artificial Intelligence in democratic political contexts.

For Han, this model is helpful to distinguish between the bits that political philosophers (or political theorists, if you prefer) ought to think about and which they can ignore. From this perspective, what is relevant for political philosophy are the input and output dimensions. Regardless of what the technology does in the process, political philosophy should focus on policies and outcomes in order to evaluate whether, for example, artificial intelligence should be used for legislative or administrative work. For his profession, Han argues, it makes most sense to focus on these two dimensions.

I think that's correct for a lot of considerations regarding democratic theory when we are thinking about the legitimacy or equality and so on — probably not for all, though.

And I don't think that this is always helpful for applied questions regarding migration infrastructure. Quite often, we need to think about the process dimension here

as well, because even with moderately good input we may get unacceptable outcomes and it may be exactly at the process level where things go pear-shaped.

## **Applying IPO**

When applying the IPO model to normative questions regarding migration infrastructure, the first step requires us to clarify what counts as input, as process, and as output. Regardless of what some people think, this is not always obvious and depends a lot on one's particular perspective.

We can think, for example, at the input level of a certain policy, or set of policies, that has been adopted by a political community in regard to handling immigration to their territory.

At the process level, we have some new software or AI product that allows officials to assess more quickly or more accurately whether an individual qualifies as refugee (based on current laws and guidelines), in which case the individual would require different treatment than migrants who are not classified as refugees.

At the outcome level we have, at least from a conservative legalistic point of view, a more just decision process that results in more genuine refugees classified as such, allowing appropriate treatment of their cases while saving resources on non-refugee migrants.

This is only a very rough sketch, of course, and you probably feel that a lot of important details are still missing. That's definitely right and I only want to use this case to show how much flexibility there is with the details if we are using IPO — and that's a flexibility we have, I think, in any other approach to complex questions, too, regardless whether they are theoretical or empirical approaches. You always have to decide what variables to track while others need to stay fixed, or decide what data points you want to compare rather than others.

With our little sketch, we could, for example, ask ourselves where to locate the operators handling the software. These are the folks who actually speak with immigrants, enter their data, and assess the replies from the system. These are people who are directly interacting with the digital migration infrastructure.

If we want to draw a strict distinction between human and technological factors, we might want to claim that operators are located at the input level and the process is only the technological process that will accept the entered data and provide some sort of output. This is, as I understand it, Han's perspective — the technology is something political philosophers should not have to deal with. It only adds unnecessary noise when considering normative political questions and likely does not fit with their field of expertise.

But we could locate the operators differently. As the people that are directly deciding what data about immigrants to enter and perhaps what data to omit, they

have an impact on how the software evaluates an immigrant's status. Further, we could also assume (hope, at least) that these operators would also have the power to intervene if the machine recommendation looks wrong or seems to ignore certain factors of an immigrant's background. In this case, operators can be understood to be located neither at the input nor at the output level but at the process level. The process, from this perspective, is not merely technical but also includes people whose actions can matter and for whom we may want to outline ethical requirements when doing their job.

Particularly with the kind of complex digital infrastructure used in migration management nowadays, it is often far from obvious along what lines to distinguish between input, process, and output. Quite often, there is not only one correct way of determining that but it rather depends on what the research question is and whether we have a valid justification for using the distinctions we choose.

## Example

### Self-Sovereign Identity for Refugees

A good case for illustrating the advantages of an IPO approach to digital migration infrastructure is the debate around *self-sovereign identity*, or *SSI*, for refugees.

As a starting point, I use an article from 2022, 'Self-Sovereignty for Refugees? The Contested Horizons of Digital Identity' by Margie Cheesman, where she provides an ethnographic study of how SSI is discussed and imagined within the aid industry (Cheesman 2022).

SSI refers to user-controlled and decentralised forms of digital identification and it is closely linked to blockchain technology. Some of you might be familiar with blockchain technology, but since probably not everyone here is, let me summarise briefly what it does.

A blockchain is essentially a shared digital ledger, that is, a record of transactions or other information that many computers hold copies of at once, rather than it sitting on a single central server. When something new is added, the network of computers collectively agrees it's valid before it gets recorded. And once recorded, it's extremely difficult to alter, because you would have to change every copy simultaneously, and every block in each of the distributed blockchain copies holds information about its preceding blocks. This design lets people who don't necessarily trust each other keep a common record that is tamper-resistant without any need for a central authority like a bank or government for oversight.

Note that not all blockchain technologies must necessarily be distributed ledger technologies, we **can** also have blockchain records held on a single, centralised server. At the same time, **not all** distributed ledgers must be based on blockchain technology. However, in most cases, blockchain applications are distributed ledger

applications, and to my understanding, that's also the type of application discusses in the context of SSI.

Blockchain is widely known for its use for Bitcoin and other cryptocurrencies, but can and has been applied to a range of different applications, e.g., supply-chain tracking or digital contracts. One could say that blockchain technology is an attempt to replace institutional trust with mathematical and procedural trust.

The idea of using blockchain in refugee administration and management processes looks quite promising at first: refugees who have lost documentation or who lack access to national identity systems could hold their own portable, cryptographically secured digital identity that can be verified through a blockchain, without depending on an unreliable or hostile government. This would grant refugees more autonomy and more privacy when they interact with potential receiving states, with aid organisations, or with other kinds of service providers.

But of course, things are not really that easy. In her paper, Cheesman identifies four areas where competing logics shape how SSI is perceived by aid industry professionals: the presumed neutrality of the technology, the assumed capacities (or incapacities) of refugees, the relationship between SSI and the nation state's monopoly on identity verification, and the economic models that emerge around digital identity data. Her central finding is that it is not at all clear, whether this sort of technology should be understood as (overall) beneficial or not. It *both* seen as a tool of refugee empowerment *and* as a means of extending existing structures of bureaucratic and commercial power over refugees.

I think that this is correct. However, Cheesman's ethnographic and discourse-oriented approach focuses on how the people who are *discussing* SSI understand and imagine the use of these technologies, rather than on the technical infrastructure that would have to *realise* it.

If we use the IPO model to see how blockchain refugee identification works as part of digital border infrastructure, we can say more about *where* the normative problems actually materialise and perhaps even can describe these problems in more detail.

## **Input**

At the input level, we find the policy frameworks and design decisions that determine what the SSI infrastructure will be working with. These include international frameworks such as UN Sustainable Development Goal 16.9 (*by 2030, provide legal identity for all, including birth registration*), UNHCR registration protocols, and the migration and asylum policies of host states. That means that there is no *single* input source but that all of these things feed into the system at input level, plus technological design specifications and the commercial interests of

corporate partners. Each of these actors contributes different requirements, assumptions, and constraints.

## **Process**

The process level is where an infrastructural perspective becomes most illuminating. The SSI infrastructure is not a single technology. It is a combination of several interconnected technical layers: biometric capture devices at the periphery, the blockchain for storing and verifying identity credentials, cryptographic key management systems, mobile wallet applications (if refugees are to hold their own data), consensus algorithms that validate transactions across the network, and interfaces to existing state and institutional databases. Each of these components involves different actors, different technical constraints, and different points where things can go wrong.

Let's look at the blockchain layer specifically. One of the properties most frequently attributed to blockchain is immutability: once data is recorded, it cannot be easily altered. In many contexts, this is a virtue because it prevents tampering and fraud. But in an identity infrastructure for refugees, immutability raises a distinctive process-level concern.

We can reasonably expect that at some point, for some people, incorrect data is entered. That may be due to operator error or a technical malfunction in the periphery device. Now, if incorrect data is entered, the infrastructure is designed to make that error permanent. Correcting the record is, also by design, extraordinarily difficult. This is a normative problem that emerges specifically from how the infrastructure processes information. It is also a problem that would remain largely invisible if we followed Han's suggestion to focus only on input and output dimensions.

The process level is also where the question of responsibility gaps becomes particularly acute. We have the distributed architecture of the blockchain itself, which is specifically designed so that no single node (and no single operator) is authoritative. We have the aid workers and health professionals who interact with the biometric capture and data entry systems. We have the consensus algorithms that validate identity claims across the network. And, if SSI works as intended, we have the refugees themselves who have to manage their own cryptographic keys and have to authorise data exchanges.

When something goes wrong and an identity is compromised, tracing the outcome back to a responsible agent involves navigating all of these layers. This is the Problem of Many Hands and some variant of algorithmic responsibility gap, all baked into this SSI infrastructure.

Cheesman rightly notes that the ideal subject of SSI is someone who is 'technologically expert, more expert than most refugees or any other people'. Looking at it

from an infrastructure perspective, we can find the point of interaction between users and SSI system where this matters the most: the key management system.

If refugees are to be genuinely self-sovereign, they must manage their own private cryptographic keys. Losing a private key means losing access to one's identity and, unlike a lost passport, there is no centralised authority that can issue a replacement.

This means that the infrastructure shifts the responsibility for a critical process-level function (identity verification and recovery) onto the least powerful actors in the system. This is not just a question of digital literacy, as it is often framed. It is a structural feature of the infrastructure that redistributes risk in ways that are not compatible with the stated goal of empowerment.

## Output

As I already mentioned before, the promises of SSI are considerable: it provides portable, privacy-enhancing, user-controlled digital identities that empower refugees with greater autonomy. However, we can distinguish between different kinds of output. There is the *technical output* of the SSI system: a verified digital credential stored on a distributed ledger. Then there is the *normative output* that matters: whether this credential actually translates into legal recognition, access to rights, and meaningful political status.

The stakeholders in Cheesman's study are largely sceptical about this translation. They are very clear that 'states are the bottom line' and that governments would likely resist accepting SSI as a substitute for state-issued identification.

The infrastructure perspective helps explaining why: SSI infrastructure and state identity infrastructure are, in most cases, not interoperable. They operate on different technical standards, different governance models, and different assumptions about authority. A blockchain-verified credential, and that's the technical output of such an SSI, is not an input that existing state systems are designed to accept. This means that the central promise of SSI of a portable identity that is recognised across borders meets a roadblock here. And, at the end of the day, states still control access to the most important forms of legal recognition, so they decide whatever concessions may be made to make these different systems more compatible.

What is more likely to emerge as a near-term output is what can be called *functional identity*. That would be a non-official digital profile that allows refugees to access some services (healthcare, financial inclusion, aid distribution) without constituting legal or foundational identity.

We should not dismiss this, it's a genuinely useful output because access to healthcare and financial services matters enormously to refugees. But it's significantly less than it says in some of the adverts.

What we also need to consider is that this functional identity exists within an infrastructure ecosystem that includes further actors with their own interests, next to states and refugees. The companies that build the infrastructure can and will likely act as intermediaries and extract commercial value from the data their infrastructure processes.

Cheesman describes scenarios where refugees could monetise their own data through SSI but also notes that this may just reproduce the dynamics of so-called platform capitalism under a different label. From an infrastructure perspective, this is fairly predictable: whoever controls critical infrastructure components occupies a position of structural power, regardless of the decentralised rhetoric.

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